

ECON 144 — FORMULA & TEST CHEAT SHEET

1. ADF TEST H_0 : unit root (nonstationary) R: `adf.test(x)`
2. Ljung-Box H_0 : No autocorrelation R: `Box.test(x, lag=10, type="Ljung-Box")`
3. ARIMA ACF: MA identification PACF: AR identification
4. ARCH/GARCH $\text{ARCH}(q)$: $\sigma^2 = \omega + \sum \alpha_i \varepsilon^2$
 $\text{GARCH}(1,1)$: $\sigma^2 = \omega + \alpha \varepsilon^2 + \beta \sigma^2$ Stationarity: $\alpha + \beta < 1$ Long-run variance: $\omega/(1-\alpha-\beta)$
5. COINTEGRATION (Engle–Granger) H_0 (A DF on residuals): no cointegration ECM: $\Delta y = \alpha + \beta \Delta x + \lambda u_{t-1}$
6. DIEBOLD–MARIANO DM = $(\text{mean}(\text{loss_diff})) / (\text{NW_se})$ H_0 : equal predictive accuracy
7. MZ REGRESSION $y = \alpha + \beta f$ H_0 : $\alpha=0, \beta=1$
8. KALMAN FILTER State: $\theta_t = \theta_{t-1} + \eta_t$
Measurement: $y_t = H\theta_t + \varepsilon_t$ Kalman Gain: $K_t = P_{\text{pred}} H'(H P_{\text{pred}} H' + R)^{-1}$
9. MARKOV CHAINS Ergodic if irreducible + aperiodic. Stationary distribution: solve $\pi = \pi P$.